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Module 4:
Basic DC networks

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1 Introduction

Electrical networks are omnipresent in today's world. They consist of interconnections of various electrical components that are linked together by connecting lines. In reality, they come in a wide variety of functions and sizes. An electrical network can be a small electronic circuit within a microcontroller measuring just a few micrometres, or an electrical power distribution network extending over several thousand kilometres, as symbolically shown in Figures 1.1 and 1.2.

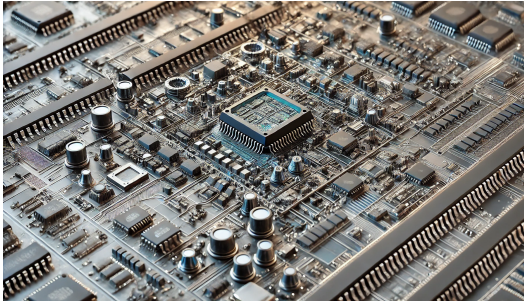


Figure 1.1: Symbolic image of a small electrical network on a circuit board



Figure 1.2: Symbolic image of a large electrical network

This chapter provides an introduction to basic calculations in direct current electrical networks. The aim of these calculations is usually to calculate the voltages and currents in all components of the network. Real components are usually simplified and represented as a combination of ideal two-pole circuits (examples: Table 2.1). The interconnection of these dipoles can be used to construct a simplified model of the real circuit, which replicates the basic behaviour of the circuit. This model forms the basis for the mathematical calculation of the circuit.

Since only linear components such as resistors or ideal DC voltage sources are used in this chapter, an analytical calculation is always possible. In advanced modules, however, non-linear components such as real operational amplifiers or diodes are introduced. In this case, an iterative or numerical approach is often necessary.

2 Two-terminal elements and reference direction systems

Learning objectives: Two-terminal elements and reference direction systems

The students can

- explain the concept of two poles, name common two poles and use their circuit symbols
- explain the generator and consumer arrow system and apply it in accordance with conventions
- explain the central elements of the basic circuit and recognise the relationships between them

2.1 Two-terminal elements

A two-pole component (Fig. 2.1) is a component with two external terminals. The internal structure of these two-pole components can vary greatly in terms of type and complexity. For example, a simple electrical resistor, but also a voltage source (e.g. a car battery) or a hairdryer

(provided it only has two connecting wires) can be regarded as a two-pole component. Components with more connecting wires are referred to as three-pole (e.g. refrigerator with protective earth conductor), four-pole (e.g. transformer) or even five-pole (electric cooker with three-phase connection, protective earth conductor and neutral conductor) components.

Regardless of its internal complexity, a two-terminal network in an electrical network can be fully characterised by the relationship between current and voltage at its connection points, known as **terminal behaviour**. It should be noted that the current I_1 of a two-terminal network shown in Figure 2.1 is always equal to the current I_2 .

The practical implementation of the components, such as actual dimensions, material properties, parasitic effects or internal inhomogeneous field strength distributions, are neglected in the network calculation with two-pole elements.

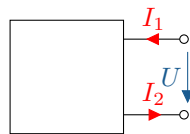


Figure 2.1: General representation of a two-terminal network with equal currents I_1 and I_2 and applied voltage U

Some of the two-pole switches already familiar from previous chapters are listed as examples in Table 2.1.

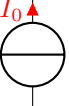
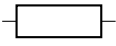
Two-terminal elements	Circuit symbol
ideal voltage source	
ideal power source	
Capacitor (capacitance)	
ohmic resistance	
Coil (inductance)	

Table 2.1: Examples of various two-pole devices and their circuit symbols

2.2 Reference direction systems

The choice of counting directions for current and voltage is essentially arbitrary. When calculating electrical networks, an attempt is often made to introduce the counting directions in such a way that the currents and voltages are positive. This makes perfect sense for quantities that are known from the outset. For unknown quantities, however, the counting direction must be determined arbitrarily. This does not mean that the current actually flows in the direction of the arrow or that a positive voltage is present in the direction of the arrow. The actual direction is then expressed by the sign of the voltage. For a sign-correct description of currents and voltages, dimensioning with counting arrows is therefore absolutely necessary.

For two-pole systems, a distinction is made between the two different counting arrow systems presented in Table 2.2:

Key point: Consumer and generator sign conventions

Passive sign convention (PSC): Current and voltage are counted in the same direction at the two poles. To be used with passive two-pole devices (e.g. resistors).

Active sign convention (GSC): Current and voltage are counted in opposite directions at the two poles. To be used with active two-pole devices (e.g. voltage sources).

Key point: Sign convention arrows

Reference arrows are used to define current and voltage sign conventions and must not be confused with vectors.

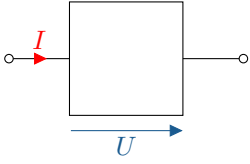
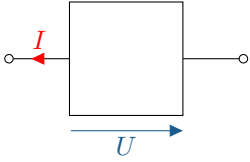
Sign convention	Generated power	Consumed power	Passive sign-convention arrows
PSC	$P = -UI$	$P = UI > 0$	
ASC	$P = UI > 0$	$P = -UI$	

Table 2.2: Comparison between the passive sign convention (PSC) and the active sign convention (ASC).

2.3 The fundamental circuit

The previously introduced two-pole (or multi-pole) systems can be combined to form an electrical network.

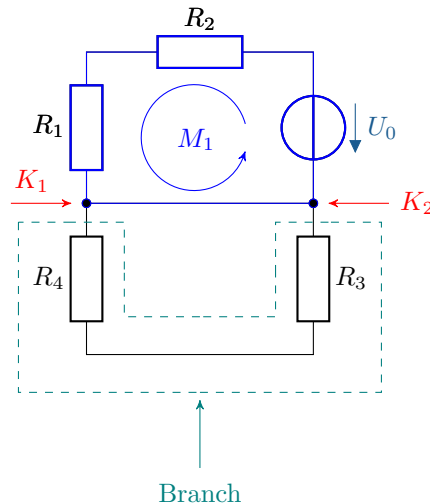


Figure 2.2: The basic electrical circuit with branches, nodes and meshes marked

The idealised two-pole **branches**, which, as shown in Fig. 2.2, can also consist of several two-pole elements connected directly in series. The same current flows through all elements of a branch. In addition to the branch marked in green in Figure 2.2, the two-pole group R_1 , R_2 and U_0 as well as the short circuit in the drawing each form another branch.

The connection points where at least three branches meet are called **nodes** or junctions. A flowing electric current can split up here onto the different branches. However, the electric potential is identical for all connected terminals. A node is marked in the circuit diagram by a filled circle and is referred to as K_n . Closed paths of at least two branches in a row within a network are called **meshes** and are abbreviated as M_n . In the circuit diagram, in addition to the designation of the mesh, a direction of circulation is often indicated by an arrow that shows the direction of circulation of the mesh. In the basic circuit drawn here, in addition to the drawn mesh M_1 consisting of R_1 , R_2 , U_0

and the short circuit between nodes K_1 and K_2 : The branch marked in green together with the short circuit between K_1 and K_2 form a mesh M_2 . Another mesh M_3 around the outside leads around the outside of the circuit and contains all the dipoles marked, but not the short circuit between K_1 and K_2 .

3 Kirchhoff's laws

The component equations introduced in the previous module, which show the relationship between voltage and current at the individual components, are not sufficient to calculate all voltages and currents within a network. Kirchhoff's rules, also known as mesh or node rules, provide the equations required for this.

Learning objectives: Kirchhoff's rules

The students can

- reproduce the key statements of Kirchhoff's rules
- apply Kirchhoff's rules to simple resistance networks

3.1 Node rule I (1. Kirchhoff rule)

The node rule states that the sum of all currents flowing into a node is identical to the sum of all currents flowing out of it:

Key point:

Sum of incoming currents = Sum of outgoing currents

Mathematically expressed, this results in the following relationship:

$$\sum I_{zu} = \sum I_{ab} \quad (3.1)$$

It should be noted that the currents are evaluated according to their counting arrow direction. Currents flowing into the node are counted as positive, currents flowing out of the node are counted as negative. If the actual direction of a current is not known in advance, the counting arrow direction can be set arbitrarily. The actual current direction is determined from the numerical value as the result of the calculation. If this is negative for a current, i.e. $I < 0$, the real current flows in the opposite direction.

Alternatively, this relationship can also be expressed by the fact that the sum of all currents in a node is equal to 0:

$$\sum_{i=1}^n = 0 \quad (3.2)$$

Applied to the example nodes in Figure ??, this results in the following equations:

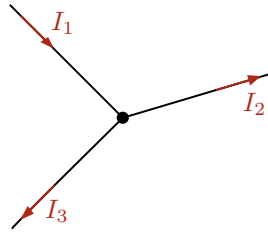


Figure 3.1: Simple example node

$$I_1 = I_2 + I_3$$

$$I_1 - I_2 - I_3 = 0$$

However, the node to which the rule refers does not have to consist of just one point. Instead, it is possible to define so-called **envelope nodes**, which completely enclose an area within a circuit. Applied to the envelope node shown in Figure 3.2 this results in the equation

$$I_1 + I_2 - I_3 - I_4 = 0$$

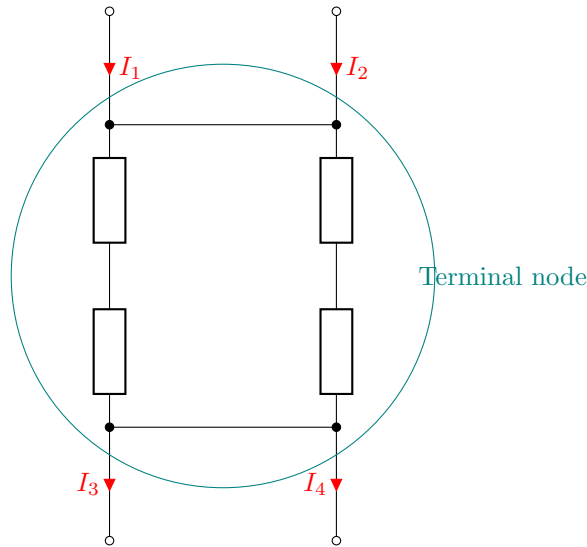


Figure 3.2: Terminal node of an electrical circuit

The node rule is not only applicable to discrete components due to the generalisation of the charge conservation law for source-free flow fields. Rather, it can be applied to any real structure. In Figure 3.3, the current flowing into the wire I is therefore equal to the total current flowing out of the envelope surface of the sheet metal cut-out.

In general, this relationship can be described by the closed surface integral over the envelope surface $\vec{A}$:

$$\iint_A \vec{J} \cdot d\vec{A} = 0 \quad (3.3)$$

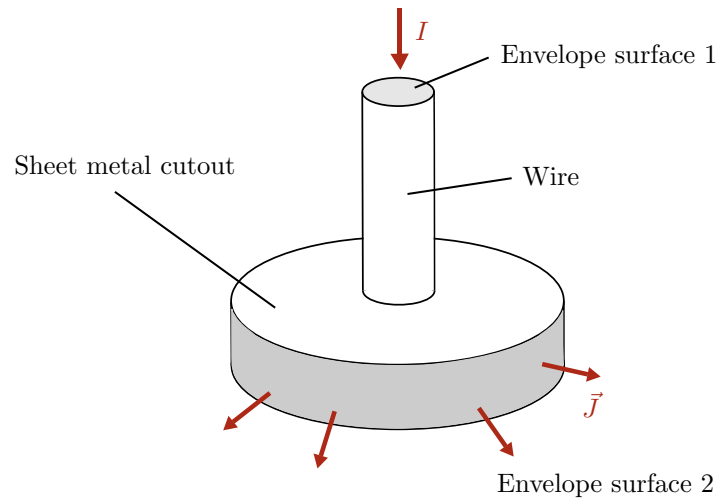


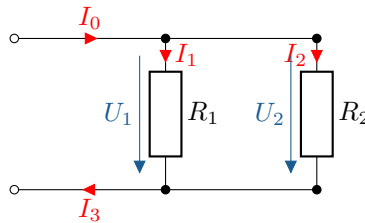
Figure 3.3: Wire connected to a sheet metal cut-out. The current I flows into the wire, while a current density $\vec{J}$ flows out of the sheet metal cut-out.

3.2 Application case node rule: Parallel connection of resistors

A basic application for the node rule is the parallel connection of resistors. The current divides at a common node, and a partial current flows through each of the individual resistors. After passing through the respective resistors, the partial currents rejoin and continue to flow as a total current. This is illustrated in example 3.1.

Example 3.1: Parallel connection of resistors

In a parallel circuit, the currents split at the common node. How large is the current I_3 in relation to the currents I_0 , I_1 and I_2 ?



$$I_0 - I_1 - I_2 = 0$$

$$\rightarrow I_0 = I_1 + I_2$$

$$-I_3 + I_1 + I_2 = 0$$

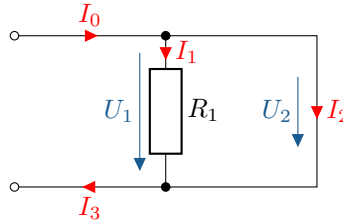
$$\rightarrow I_1 + I_2 = I_3$$

$$\rightarrow I_3 = I_0$$

If one of the resistors is replaced by an ideally conductive connection (conductance approaches infinity, resistance consequently approaches 0), the current flow in the area between the two nodes changes (example 3.2)..

Example 3.2: Parallel connection with conductor

A resistor is replaced by a conductive connection. How large are the currents I_1 and I_2 ?



$$\begin{aligned}
 I_0 &= I_1 + I_2 \\
 \text{mit } U_1 &= R_1 \cdot I_1 \\
 U_2 &= R_2 \cdot I_2 = 0 \stackrel{!}{=} U_1 \\
 &\rightarrow I_1 = 0 \\
 &\rightarrow I_0 = I_2 = I_3
 \end{aligned}$$

3.3 Mesh rule (Kirchhoff's second law)

Kirchhoff's second law (mesh rule) states that the **sum of all voltages** in a mesh is **zero**. Analogous to the direction of the currents, the direction of the individual partial voltages must also be taken into account here. If the direction arrow of a partial voltage points in the opposite direction to the direction of circulation of the mesh, then this partial voltage must be given a negative sign. If the direction of an applied voltage is not known, then an arbitrary counting arrow direction can also be assumed here. An oppositely applied voltage is also expressed in calculations with a negative sign.

Key point: Mesh rule

$$\sum_{k=1}^n U_k = 0 \quad (3.4)$$

Equivalent to the above definition, it can be stated that the sum of all voltages connected in the same direction at voltage sources corresponds to the voltage that drops across the consumers.

Key point: Mesh rule 2

Total voltage at voltage sources = total voltage at consumers

Applied to the example network shown in Figure 3.4, the following mesh equations result:

$$U_1 + U_2 - U_3 - U_4 = 0$$

respectively

$$U_1 + U_2 = U_3 + U_4$$

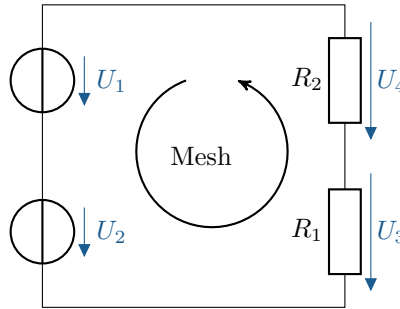


Figure 3.4: Example network consisting of two voltage sources and two resistors

The mesh equation also applies if additional currents are fed into the mesh, or if individual two-poles are passed through several times during the circulation around a closed mesh.

The general relationship, whereby any integrated voltage along a closed contour equals 0, can be described as follows according to Maxwell's equations:

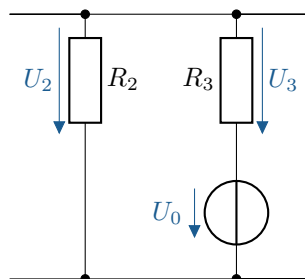
$$\oint_s \vec{E} d\vec{s} = 0 \quad (3.5)$$

3.4 Application case mesh rule: Series connection of resistors

While the nodes in electrical networks are primarily important for current, the branches and thus also the meshes are primarily of interest for calculating voltages. In a series connection of resistors in a branch, all partial voltages add up to a total voltage with the correct sign. An application case for determining a partial voltage is shown in Example 3.3.

Example 3.3: Series connections in networks

In a series connection, the voltages add up to a total voltage. How large is the voltage U_0 ?



Set up the mesh counterclockwise:

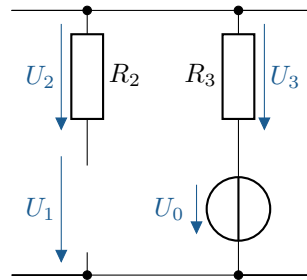
$$U_2 - U_0 - U_3 = 0$$

$$\rightarrow U_0 = U_2 - U_3$$

The mesh rule can also be applied if the mesh to which it is applied has an interruption point (see example 3.4).

Example 3.4: Series connections with interruption point

A connection is interrupted. How large is the voltage U_1 at the point of interruption?



Set up the mesh counterclockwise:

$$U_2 + U_1 - U_0 - U_3 = 0$$

Determine the voltage U_2 using Ohm's law:

$$U_2 = I_2 \cdot R_2$$

$$\rightarrow I_2 = 0$$

$$\rightarrow U_2 = 0$$

Insert into the initial equation and solve for U_1 :

$$U_1 = U_0 + U_3$$

4 Simple resistor networks

Electrical networks are often composed of simpler sub-circuits. It is often helpful to identify these sub-circuits, simplify them, and then combine them into the overall circuit. Such combinations of components are generally permissible, provided that the **terminal behaviour**, i.e. the behaviour between current and voltage between the connection points, does not change.

Learning objectives: Simple resistance networks

The students can

- Identify subcircuits in direct current networks
- Simplify and summarise resistance networks
- Determine short-circuit and no-load data
- Apply superposition methods

4.1 Series connection of resistors

In a series connection, the same current I flows through all components (see Figure 4.1).

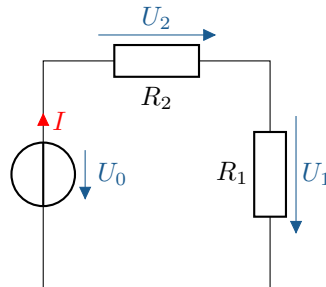


Figure 4.1: Series connection of two resistors and a voltage source

Applying the mesh rule (here: counterclockwise direction of rotation) leads to the following mesh equation:

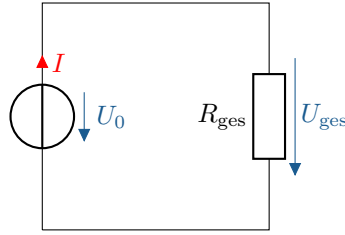
$$U_0 - U_1 - U_2 = 0$$

With the help of Ohm's law, the partial voltages U_1 and U_2 can be represented as the product of current and resistance:

$$U_0 - R_1 \cdot I - R_2 \cdot I = 0$$

$$U_0 - (R_1 + R_2) \cdot I = 0$$

In this example, resistors R_1 and R_2 can be combined by adding their resistance values to $R_1 + R_2 = R_{\text{total}}$ (Figure 4.2).

Figure 4.2: Combining R_1 and R_2 to form R_{total}

In general, the total voltage U_{tot} can be summarised according to this principle as follows:

$$U_{\text{tot}} = \sum_{k=1}^n U_k = \sum_{k=1}^n R_k \cdot I = R_{\text{tot}} \cdot I \quad (4.1)$$

A comparison of the coefficients of the last two terms of equation 4.1 yields the generally valid result for combining resistors in a series connection:

Key point: Total resistance of a series connection

$$R_{\text{tot}} = \sum_{k=1}^n R_k \quad (4.2)$$

If, instead of the resistance values R , we consider the conductance values $G = 1/R$ of the resistors, the total conductance of a series connection is:

Key point: Total conductance of a series connection

$$\frac{1}{G_{\text{tot}}} = \sum_{k=1}^n \frac{1}{G_k} \quad (4.3)$$

4.2 Parallel connection of resistors

In a parallel connection of resistors, the same voltage is applied to all components (see Figure 4.3):

$$U_1 = U_2 = U_0$$

By applying the node rule, it can be shown that the total current I_0 before the resistors is divided into the partial currents I_1 and I_2 :

$$I_0 - I_1 - I_2 = 0 \quad (4.4)$$

As with series connection in Section 4.1, it is often advantageous to combine resistors connected in parallel into a total resistance R_{total} (see Figure 4.4):

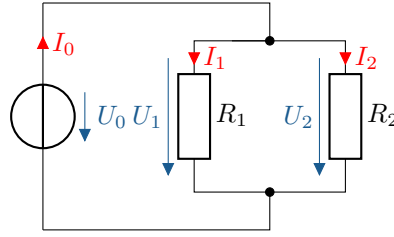
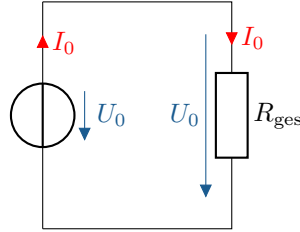


Figure 4.3: Parallel connection of two resistors and a voltage source

Figure 4.4: Summary of the parallel-connected resistors R_1 and R_2 from Figure 4.3 to R_{total}

With the help of Ohm's law, the unknown partial currents I_1 and I_2 can be quantified from the node rule (equation 4.4):

$$I_1 = \frac{U_0}{R_1}, \quad I_2 = \frac{U_0}{R_2}$$

Inserted into equation 4.4, this gives:

$$I_0 - \frac{U_0}{R_1} - \frac{U_0}{R_2} = 0$$

$$I_0 - \left(\frac{1}{R_1} + \frac{1}{R_2} \right) \cdot U_0 = 0$$

$$\rightarrow I_0 = U_0 \cdot \left(\frac{1}{R_1} + \frac{1}{R_2} \right)$$

A coefficient comparison with Ohm's law applied to Figure 4.4 ($I_0 = U_0 \cdot \frac{1}{R_{\text{tot}}}$)

provides for R_{total} in the case shown of two resistors connected in parallel:

$$\frac{1}{R_{\text{tot}}} = \frac{1}{R_1} + \frac{1}{R_2}$$

Multiplying out this expression gives the form frequently used in calculations:

$$R_{\text{tot}} = \frac{R_1 \cdot R_2}{R_1 + R_2} \quad (4.5)$$

The total resistance of a parallel connection of any number of resistors can be determined using the general formula for parallel connections:

Key point: Total resistance of a parallel connection

$$\frac{1}{R_{\text{tot}}} = \sum_{k=1}^n \frac{1}{R_k} \quad (4.6)$$

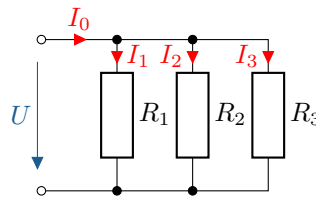
Since electrical conductance is described by the reciprocal of resistance, $G = 1/R$, the total conductance of a parallel connection of resistors is often easier to calculate:

Key point: Total conductance of a parallel connection

$$G_{\text{tot}} = \sum_{k=1}^n G_k \quad (4.7)$$

Example 4.1: Parallel connection of resistors

The three resistors $R_1 = 1 \text{ k}\Omega$, $R_2 = 10 \text{ k}\Omega$, $R_3 = 100 \text{ k}\Omega$ are connected in parallel as shown in the following figure. What is the total resistance R_{total} of this parallel connection?



Set up the equation for R_{total} according to equation 4.6:

$$R_{\text{tot}} = \frac{1}{\sum_n \frac{1}{R_n}} = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}}$$

Inserting the numerical values and solving using a calculator:

$$R_{\text{tot}} = \frac{1}{\frac{1}{1 \cdot 10^3 \Omega} + \frac{1}{10 \cdot 10^3 \Omega} + \frac{1}{100 \cdot 10^3 \Omega}} \\ \rightarrow R_{\text{tot}} = 900,9 \Omega$$

4.3 Voltage divider on resistors

One application of connecting two resistors in series is the voltage divider. It is used to divide an input voltage U_0 into two smaller partial voltages, thus providing a precisely defined output voltage U_2 .

4.3.1 The unbalanced voltage divider

First, the voltage divider shown in Figure 4.5 is unloaded, i.e. no load resistor is connected to the open terminals. All components are traversed by the same current I :

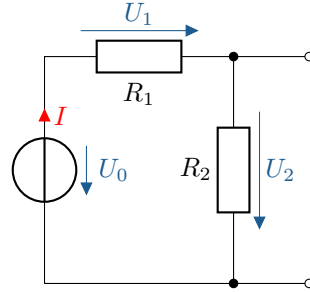


Figure 4.5: Unloaded voltage divider

$$I = \frac{U_1}{R_1} = \frac{U_2}{R_2}$$

By combining the series connection of resistors, the following result is obtained:

$$I = \frac{U_0}{R_1 + R_2}$$

Equating the two voltage terms yields the ratio equation:

$$\frac{U_2}{R_2} = \frac{U_0}{R_1 + R_2}$$

Isolating the output voltage on the left side of the equal sign ultimately leads to the general equation for the unloaded voltage divider:

Key point: Unloaded voltage divider

$$U_2 = U_0 \cdot \frac{R_2}{R_1 + R_2} \quad (4.8)$$

Example 4.2: Unloaded voltage divider

An input voltage of $U_0 = 24 \text{ V}$ is to be reduced to $U_2 = 6 \text{ V}$ using a voltage divider. What should the ratio of the resistances R_2 to R_1 be? How large should the resistances R_1 and R_2 be if the total current is to be $I = 10 \text{ mA}$?

$$\frac{U_2}{U_0} = \frac{6 \text{ V}}{24 \text{ V}} = 0,25 = \frac{R_2}{R_1 + R_2}$$

$$\rightarrow R_2 = 0,25(R_1 + R_2)$$

$$0,75 R_2 = 0,25 R_1$$

$$R_1 = 3 R_2$$

The resistance R_1 must therefore be selected to be three times greater than the resistance R_2 .

$$I = \frac{U_2}{R_2} \rightarrow R_2 = \frac{U_2}{I} = \frac{5 \text{ V}}{0.01 \text{ A}} = 500 \Omega$$

$$R_1 = 3 \cdot R_2 = 3 \cdot 500 \Omega = 1500 \Omega$$

4.3.2 The loaded voltage divider

If the output side of the voltage divider is loaded as shown in Figure 4.6, i.e. a load resistor R_L is added, then - from the perspective of the voltage source - this results in a parallel connection of the resistors R_2 and R_L .

If the equivalent resistance R_{new} of this parallel connection is determined via $\frac{1}{R_{\text{new}}} = \frac{1}{R_2} + \frac{1}{R_L}$, the ratio of the resistances of the voltage divider changes, and consequently also the output voltage. This change must be taken into account when designing a loaded voltage divider.

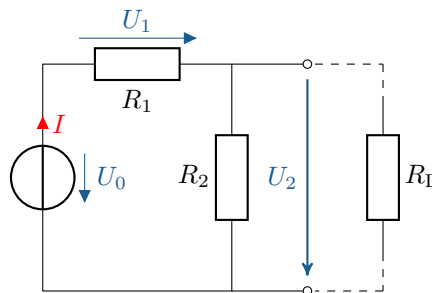


Figure 4.6: Voltage divider loaded with resistance R_L

Example 4.3: Loaded voltage divider

A voltage divider is designed to reduce an input voltage of $U_0 = 20 \text{ V}$ to an output voltage of $U_2 = 5 \text{ V}$. The voltage divider is loaded with $R_L = 200 \Omega$. The resistance $R_1 = 300 \Omega$ is specified.

How large must R_2 be in order to achieve the desired output voltage?

First, the required total value R_{new} of the parallel connection of R_L and R_2 must be calculated:

$$U_2 = U_0 \cdot \frac{R_{\text{neu}}}{R_1 + R_{\text{neu}}}$$

$$R_{\text{neu}} = \frac{5 \text{ V}}{20 \text{ V}} \cdot (R_1 + R_{\text{neu}})$$

$$0.75 \cdot R_{\text{neu}} = 0.25 \cdot R_1 = 75 \Omega$$

$$\rightarrow R_{\text{neu}} = 100 \Omega$$

The resistance R_1 can be calculated using the formula for parallel circuits:

$$100 \Omega = \frac{R_L \cdot R_2}{R_L + R_2} = \frac{200 \Omega \cdot R_2}{200 \Omega + R_2}$$

$$R_2 \cdot \frac{200 \Omega}{100 \Omega} = 2 \cdot R_2 = 200 \Omega + R_2 \quad (4.9)$$

$$\rightarrow R_2 = 200 \Omega$$

4.4 Current dividers in resistor networks

A parallel connection of two resistors, as shown in Figure ??, divides a current I_0 flowing in a circuit into two partial currents I_1 and I_2 . Such a circuit is called a current divider. The current I_0 flowing in the circuit is divided by the resistors R_1 and R_2 according to the formula:

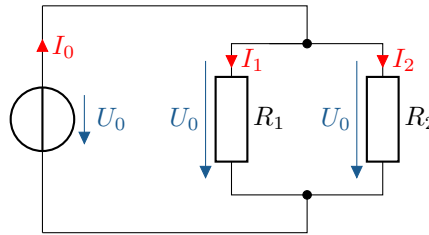


Figure 4.7: Current divider of current strength I_0 into partial current strengths I_1 and I_2 , consisting of resistors R_1 and R_2 connected in parallel.

All components of this circuit have the identical output voltage U_0 . Using Ohm's law, this voltage across resistors R_1 and R_2 can also be described as the product of the resistance value and the current:

$$U_0 = I_1 \cdot R_1 = I_2 \cdot R_2$$

By combining resistors R_1 and R_2 to form R_{total} as described in Figure 4.4, the total current I_0 can be determined as a function of the output voltage U_0 :

$$U_0 = I_0 \cdot R_{\text{tot}}$$

Equating the two current terms yields the following relationship between I_0 and I_2

$$I_2 \cdot R_2 = I_0 \cdot R_{\text{tot}}$$

The parallel connection R_{total} can be calculated directly from the two resistance values R_1 and R_2 as shown in equation 4.5:

$$I_2 \cdot R_2 = I_0 \cdot \frac{R_1 \cdot R_2}{R_1 + R_2}$$

The reduction of R_2 results in the division ratio of the electric current at the current divider:

Key point: Stromteiler einer Parallelschaltung

$$I_2 = I_0 \cdot \frac{R_1}{R_1 + R_2} \quad (4.10)$$

4.5 Special operating conditions for active two-pole systems

Active two-pole circuits can be grouped together as **equivalent voltage sources** or **equivalent current sources**. This can be particularly useful for simplifying electrical networks.

4.5.1 Short-circuit and no-load test on the replacement voltage source

The real source of tension

As already known from Module 3 *Electrical Components*, a **real voltage source** consists of an ideal voltage source U_0 and an internal resistance R_i .

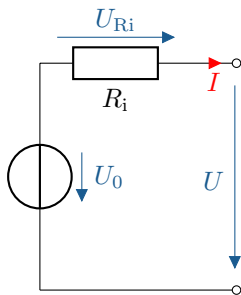


Figure 4.8: Real voltage source consisting of an ideal voltage source U_0 and an internal resistance R_i

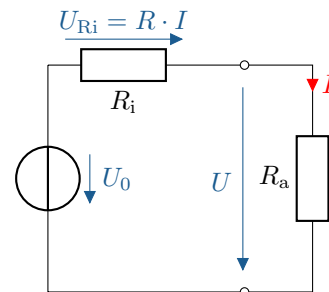


Figure 4.9: Real voltage source loaded with a load resistance R_a

If the real voltage source is unloaded, i.e. no component is connected to the output terminals, no current I flows (see Figure 4.8).

$$I = 0$$

$$\rightarrow U_{Ri} = R_i \cdot I = 0$$

Since the voltage across the internal resistance R_i is 0 volts, the following applies to the terminal voltage U :

$$U = U_0$$

If the real voltage source is loaded with a resistor R_a as shown in Figure 4.9, a current $I > 0$ flows through the line.

Consequently, a voltage $U_{R_i} = R_i \cdot I$ drops across the internal resistance R_i , and the terminal voltage is

$$U = U_0 - U_{R_i}$$

To reduce the voltage drop within the voltage source, the internal resistance R_i should be kept as low as possible:

$$U_0 \ll I \cdot R_i$$

The replacement voltage source

Each active two-terminal network with at least one source and any number of resistors can be combined into a equivalent voltage source (Figure 4.10). As with a real voltage source, it is sufficient to determine an ideal voltage source U_0 and a combined internal resistance R_i to fully characterise the equivalent voltage source. Externally, it has completely equivalent terminal behaviour to the original circuit at its two connection terminals.

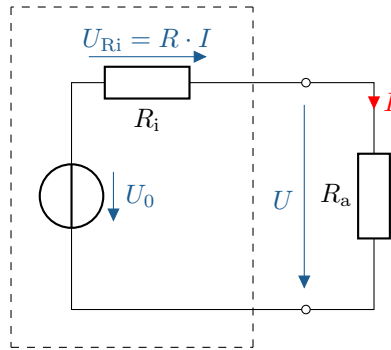


Figure 4.10: The equivalent voltage source (dashed box) has the same equivalent circuit diagram as a real voltage source.

As with a loaded real voltage source, the resulting output voltage U depends on both the internal resistance R_i and the load current I (Figure 4.11):

$$U = U_0 - I \cdot R_i$$

The parameters U_0 and R_i required for characterisation can be determined using two tests: the short-circuit test and the no-load test.

In the no-load test, the two terminals of the equivalent voltage source are left open (Figure 4.12).

Da durch die offenen Klemmen der Strom I durch den Widerstand R_i null Ampere beträgt, fällt auch keine Spannung über ihn ab. Die Leerlaufspannung ist also mit der Spannung U_0 identisch:

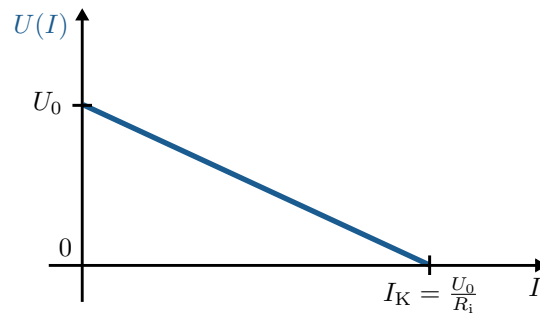


Figure 4.11: The voltage U applied to the terminals of the replacement voltage source as a function of the resulting current I

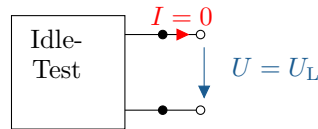


Figure 4.12: Determination of the open-circuit voltage U_L of an equivalent voltage source

Key point: Open-circuit voltage of an equivalent voltage source

$$U_0 = U_L$$

In the short-circuit test, the two terminals of the active two-pole are directly short-circuited, i.e. an ideally conductive connection is established between them (Figure 4.13).

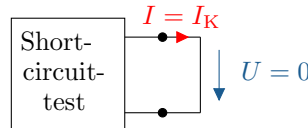


Figure 4.13: Determination of the short-circuit current I_K of an equivalent voltage source

No output voltage U can be applied across the short circuit thus created:

$$U = 0$$

Consequently, the entire voltage generated by U_0 drops across the internal resistance R_i . According to Ohm's law, this determines the short-circuit current I_K .

$$U_{Ri} = U_L = R_{Ri} \cdot I_K$$

The short-circuit current I_K is therefore:

$$I_K = \frac{U_L}{R_i}$$

The internal resistance R_i can therefore be calculated as follows:



Key point: Internal resistance of an equivalent voltage source

$$R_i = \frac{U_L}{I_K}$$

Attention: U_L und I_K are certain quantities in two completely independent experiments that do not occur simultaneously. The formula therefore only looks like Ohm's law at resistance R_i , but it is not.

4.5.2 Short-circuit and no-load test on the emergency power source

The real power source

A **real current source** also consists of an ideal current source I_0 and an internal resistance R_i (see Figure 4.14). However, in contrast to the real voltage source, this is connected in parallel rather than in series with the source.

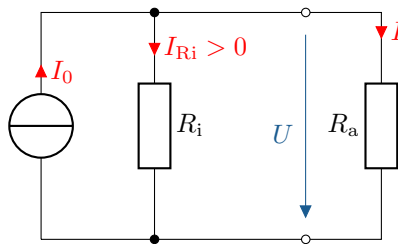


Figure 4.14: Real current source consisting of an ideal current source I_0 and an internal resistance R_i , which is loaded with a resistance R_a

Depending on the output voltage U that occurs, the current $I_{Ri} = U/R_{Ri}$ flows through the finite internal resistance R_i , so the output current I is given by



Key point: Output current of a real current source

$$I = I_0 - \frac{U}{R_i}$$

In order to reduce the output current I through the internal resistance, this should be selected as large as possible:

$$I_0 \gg \frac{U}{R_i}$$

The backup power source

As an alternative to the replacement voltage source, an active two-pole can also be represented as a replacement current source (Figure 4.15). It can also be fully characterised by an ideal current source I_0 and a combined internal resistance R_i .

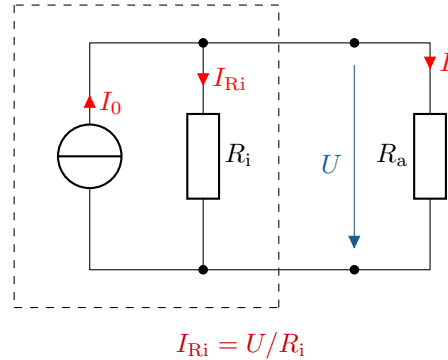


Figure 4.15: The replacement current source (dashed box) has the same equivalent circuit diagram as a real current source.

Similar to the equivalent voltage source, the parameters I_0 and R_i can also be determined here using the open-circuit and short-circuit tests.

In the **open-circuit test**, the entire current I_0 flows through the internal resistance. The following open-circuit voltage U_L is established at the terminals connected in parallel:

$$U_L = R_i \cdot I_0$$

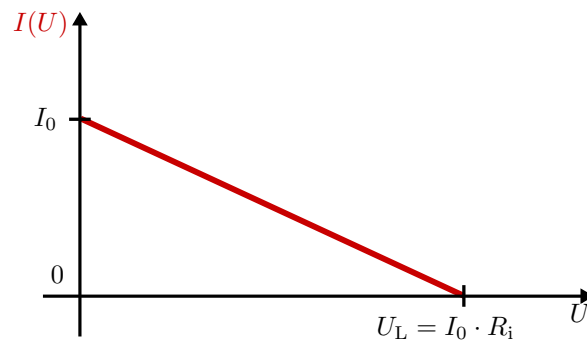


Figure 4.16: The current I flowing out of the backup power source as a function of the voltage U

In the **short-circuit experiment**, the two terminals of the equivalent current source are short-circuited. Since the voltage across the short circuit and thus also across the internal resistance connected in parallel is zero volts, no current flows through R_i .

The entire current I_0 therefore flows through the short circuit:



Key point: Short-circuit current of a backup power source

$$I_0 = I_K$$

With $U_L = R_i \cdot I_0$, the internal resistance can now be determined:

Key point: Internal resistance of an equivalent current source

$$R_i = \frac{U_L}{I_K}$$

4.5.3 Conversion of voltage and current sources

In network analysis, it can be useful to convert current sources into voltage sources or voltage sources into current sources. This works identically for real current and voltage sources, as well as for equivalent current and voltage sources. The value of the internal resistance R_i remains the same, but its position in the equivalent circuit diagram changes (see Figure ??)

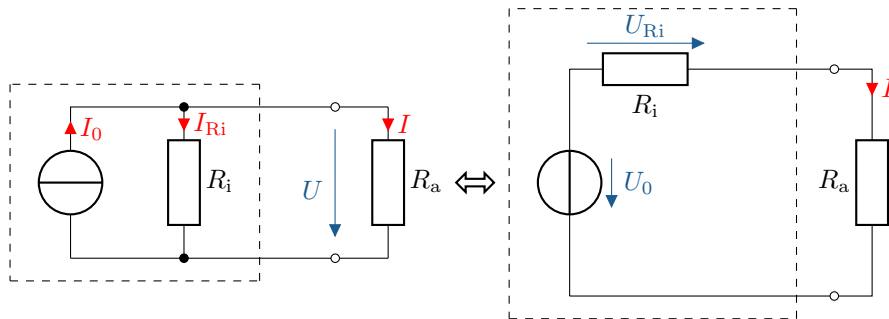


Figure 4.17: Conversion of a backup power source (left) into a backup voltage source (right) and vice versa

The open-circuit voltage and short-circuit current remain unchanged during the conversion of the source and are converted to the other source.

Attention: The Sign Conventions of current and voltage sources converted into each other are directed in opposite directions!

When converting a current source into a voltage source, the voltage U_0 is:

Key point: Voltage of a converted voltage source

$$U_0 = I_0 \cdot R_i$$

The current I_0 when converting a voltage source into a current source is:

Key point: Current from a converted current source

$$I_0 = \frac{U_0}{R_i}$$

5 Simple capacitor networks

Like resistors, capacitors can also be connected to form networks. The simple basic circuits, which can in turn be interconnected to form networks of any complexity, are presented below.

5.1 Parallel connection of capacities

If several capacitances are connected in parallel in a circuit, they can be combined to form a total capacitance (see Figure ??).

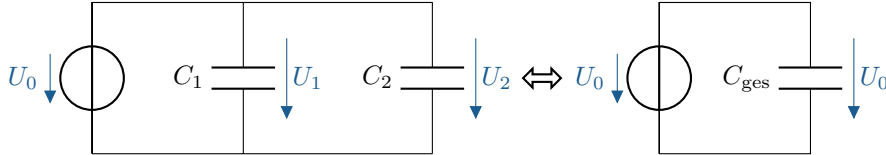


Figure 5.1: ZCombining two capacitances C_1 and C_2 connected in parallel to form a total capacitance C_{total}

The same voltage is applied to all individual capacities:

$$U_0 = U_1 = U_2$$

The charge stored on each of the individual capacitors can be determined as follows:

$$Q_k = C_k \cdot U_0$$

The total charge C_{total} on the equivalent circuit diagram in Figure ?? corresponds to

$$Q_{\text{ges}} = C_{\text{ges}} \cdot U_0$$

The total load is made up of the sum of the individual loads:

$$Q_{\text{ges}} = Q_1 + Q_2 = C_1 \cdot U_0 + C_2 \cdot U_0$$

$$Q_{\text{ges}} = (C_1 + C_2) \cdot U_0$$

A comparison of the quotient with the original calculation of the total charge Q_{total} now shows:

$$\rightarrow C_1 + C_2 = C_{\text{ges}}$$

Even for any number of capacities connected in parallel, the sum of the individual capacities equals the total capacity:



Key point: Total capacity of a parallel connection

$$C_{\text{ges}} = \sum_{k=1}^n C_k$$

5.2 Series connection of capacitances

When capacitors are connected in series, the total voltage U_0 applied to the input terminals is divided into partial voltages U_k across the individual capacitors (Figure ??).

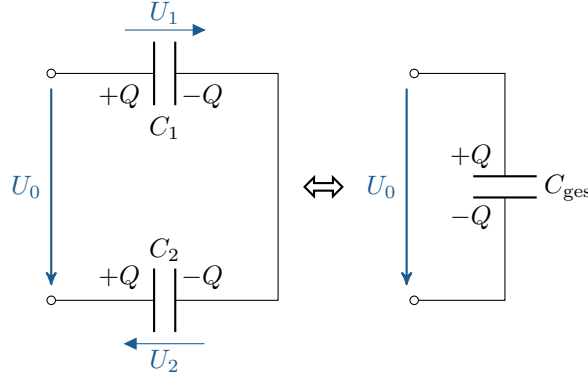


Figure 5.2: Combining two capacitances C_1 and C_2 connected in series to form a total capacitance C_{total}

A charge of $\pm Q$ is applied to the capacitor plates connected to the terminals. Both plates of the same capacitor always have charges of equal magnitude, which means that both capacitors at the terminals of the series connection have a charge of $\pm Q$ on both plates. The outer plate of the next connected capacitor must have an identical charge with the opposite sign, as no charge carriers can escape or be added to the originally electrically neutral connection. Following this pattern, all capacitors in a series connection have the same charge Q .

Based on the basic capacitor equation $Q = C \cdot U$, the partial voltages across the capacitors shown in Figure ?? are:

$$U_1 = \frac{Q}{C_1}, \quad U_2 = \frac{Q}{C_2}$$

Inserted into the result of the mesh cycle of the output circuit, this gives:

$$U_0 = U_1 + U_2 = \left(\frac{1}{C_1} + \frac{1}{C_2} \right) \cdot Q$$

The voltage U_0 in the equivalent circuit diagram can be calculated as follows:

$$U_0 = \frac{1}{C_{\text{tot}}} \cdot Q$$

A comparison of ratios reveals:

$$\frac{1}{C_1} + \frac{1}{C_2} = \frac{1}{C_{\text{tot}}}$$

In general, the following applies to a series connection of any number of capacitors:

Key point: Total capacity of a series connection:

$$\frac{1}{C_{\text{tot}}} = \sum_{i=1}^n \frac{1}{C_i} \quad (5.1)$$

For two capacitors connected in series, the total capacitance C_{total} can also be determined using the following expression, analogous to the parallel connection of resistors:

$$C_{\text{tot}} = \frac{C_1 \cdot C_2}{C_1 + C_2}$$

5.3 Voltage divider on capacitances

Similar to the voltage divider with two resistors connected in series, the ratio of the divided voltage can also be determined for two capacitors with identical charges connected in series (see Figure ??).

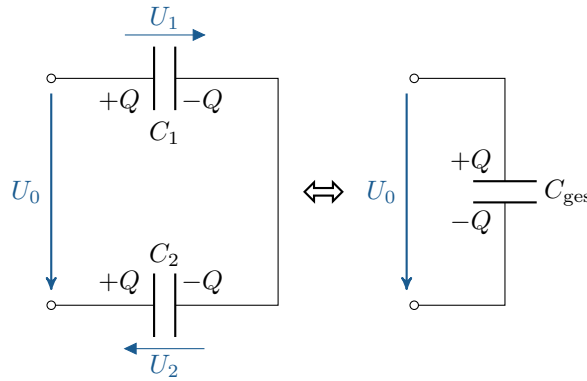


Figure 5.3: Voltage divider in a series connection of two capacitances (left) and equivalent circuit diagram after combining the capacitances to C_{total}

The voltage U_1 can be determined as follows:

$$U_1 = \frac{Q}{C_1}$$

The charge Q is unknown, but it is identical for both capacitors C_1 and C_2 as well as in the equivalent circuit at C_{total} :

$$Q = C_{\text{tot}} \cdot U_0 = \frac{C_1 \cdot C_2}{C_1 + C_2} \cdot U_0$$

Substituting, we obtain for U_1 :

$$U_1 = \frac{1}{\cancel{C_1}} \cdot \frac{\cancel{C_1} \cdot C_2}{C_1 + C_2} \cdot U_0 = U_0 \cdot \frac{C_2}{C_1 + C_2}$$

The voltage divider for U_2 is calculated analogously:

Key point:

$$U_1 = \frac{C_2}{C_1 + C_2} \cdot U_0, \quad U_2 = \frac{C_1}{C_1 + C_2} \cdot U_0$$

6 Measuring current and voltage

6.1 Voltage measurement in a direct current circuit

The electrical voltage that drops across a component or across an interconnection of several components can be determined using a voltage measuring device (also called a voltmeter). To do this, the voltmeter must be connected **in parallel** to this component. It is usually not necessary to interrupt the circuit to do this.

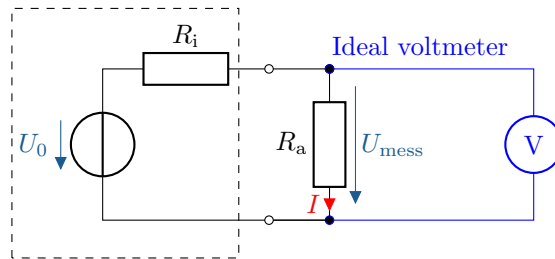


Figure 6.1: A real voltage source (box on the left) feeds a load resistor R_a . The voltage drop across it is measured with an ideal voltmeter (blue).

Key point: Internal resistance of ideal voltmeter

An ideal voltmeter (see Figure ??) has an infinitely high internal resistance.

Consequently, no current „flows past the load resistor“ through the measuring device, and the voltage measurement is performed completely **without feedback**.

A real voltmeter has a finite internal resistance R_{iV} , which is connected in parallel to the voltmeter (see Figure ??). The current flowing through this internal resistance distorts the measurement result, which is why it should be as high as possible. In modern, electronic voltage measuring devices, it is usually in the range of $10^6 \Omega$ to $10^9 \Omega$.

The voltage U_{corr} , corrected for the measurement error caused by the internal resistance R_{iV} , can be calculated as follows, given the internal resistance of the voltage source R_i :

$$U_{\text{corr}} = U \cdot \left(1 + \frac{R_i || R_a}{R_{iV}} \right)$$

6.2 Current measurement in a direct current circuit

The current I flowing through a component can be determined using a current measuring device (also called an ammeter). In contrast to voltage measurement, the ammeter must be connected **in**

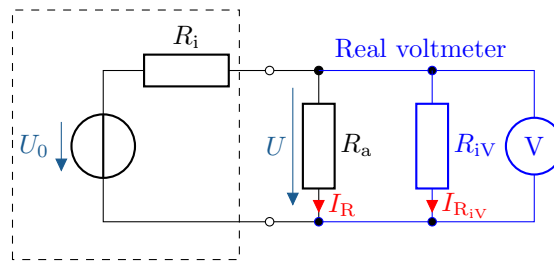


Figure 6.2: Messung der Spannung U über einem Lastwiderstand R_a durch ein reales Voltmeter mit endlichem Innenwiderstand R_{iV}

series to the component. This requires an interruption of the circuit.

Key point: INominal resistance ideal ammeter

An ideal ammeter (see Figure ??) has no internal resistance.

Consequently, the current can flow through it without any feedback.

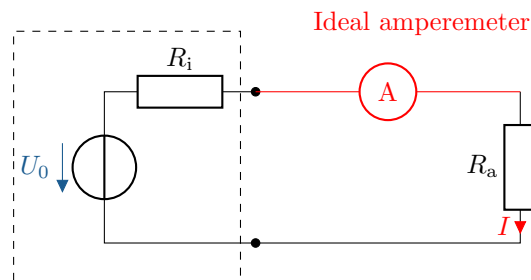


Figure 6.3: A real voltage source (box on the left) supplies a load resistor R_a . The current I flowing through the load resistor is measured by an ideal ammeter (red).

A **real ammeter** has an internal resistance R_{iA} connected in series (Figure 6.4). Here, too, the measurement result is influenced by this internal resistance. Depending on the measuring range, this typically ranges from a few $\mu\Omega$ to less than $m\Omega$ in electronic ammeters.

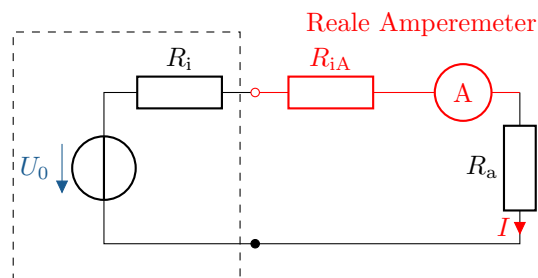


Figure 6.4: Measurement of the current I through a load resistor R_a using a real ammeter with internal resistance R_{iA} (red)

Knowing the internal resistances, the measurement error can also be calculated for the ammeter. The corrected current I_{corr} , for which the influence of the measuring device is calculated out, can be determined using the following formula:

$$I_{\text{corr}} = I \cdot \left(1 + \frac{R_{iA}}{R_i + R_a} \right)$$

6.3 Correct measurement of current and voltage

If both the voltage and the current are to be measured in a circuit, the internal resistances of the actual measuring instruments influence each other. Depending on the quality of the available measuring instruments and the prioritisation of the measured variables among themselves, a distinction is made between current-correct and voltage-correct measurement.

6.3.1 Correct measurement of electricity

In **current-correct** measurement (Figure 6.5), also known as voltage error circuitry, greater emphasis is placed on accurately determining the current. In this circuit, the ammeter displays the current flowing through the load resistor, but the voltmeter measures the sum of the voltage drops across the ammeter and the load resistor:

$$U_{\text{meas}} = U + U_{\text{Amperemeter}}$$

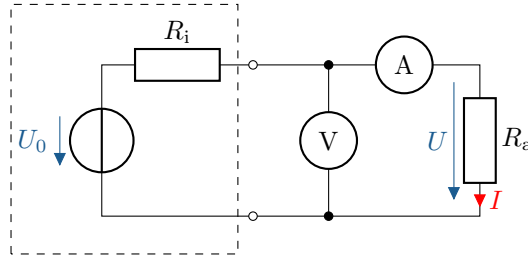


Figure 6.5: Current-correct measuring circuit for determining the current I and the voltage U at a load resistor R_a

The voltage fault circuit tends to be suitable for circuits with relatively high load resistances. The following inequality is often used as a rule of thumb for the use of this measuring circuit:

$$R_{iA} \ll R_a$$

6.3.2 Correct voltage measurement

In contrast to the current-correct measurement circuit, the voltmeter in the voltage-correct measurement circuit measures the correct voltage drop across the resistor (see Figure ??). The ammeter, on the other hand, measures the current flowing through the parallel connection of the voltmeter and load resistor. This measurement arrangement is recommended if greater emphasis is to be placed on the correct measurement of the voltage, or if the following applies to the resistors:

$$R_{iV} \gg R_a$$

If it is unclear whether current-correct or voltage-correct measurement should be used, the following rule of thumb can be applied:

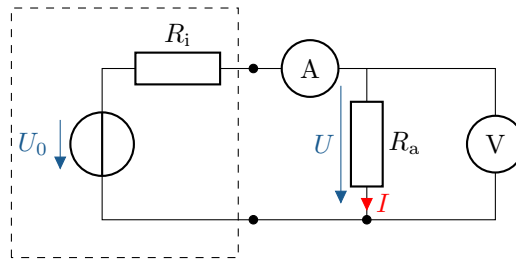


Figure 6.6: Current-correct measuring circuit for determining the current I and the voltage U at a load resistor R_a



Key point: Rule of thumb for correct current/voltage measurement

- Correct current measurement: $R_a > \sqrt{R_{iA} \cdot R_{iV}}$
- Correct voltage measurement: $R_a < \sqrt{R_{iA} \cdot R_{iV}}$

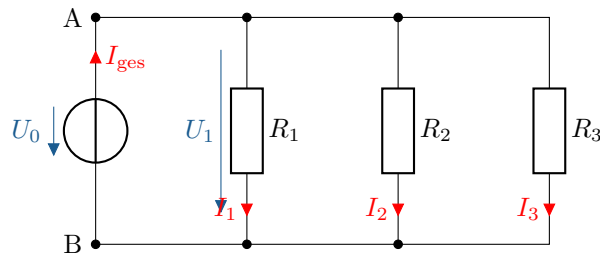
A Exercise tasks

A.1 Kirchhoff's laws

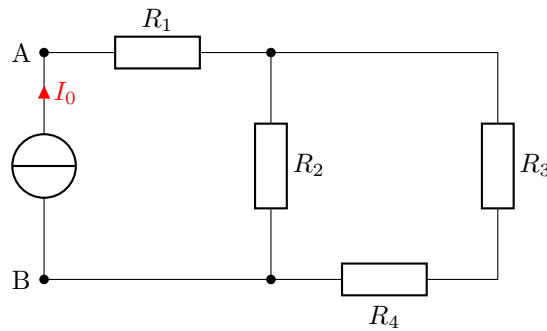
Calculate the currents, voltages and power at the resistors for the circuits. Draw all current and voltage arrows for your results. The following values are given:

$$U_0 = 12 \text{ V}; I_0 = 3 \text{ A}; R_1 = 2 \text{ } \Omega; R_2 = 4 \text{ } \Omega; R_3 = 3 \text{ } \Omega; R_4 = 3 \text{ } \Omega$$

a)



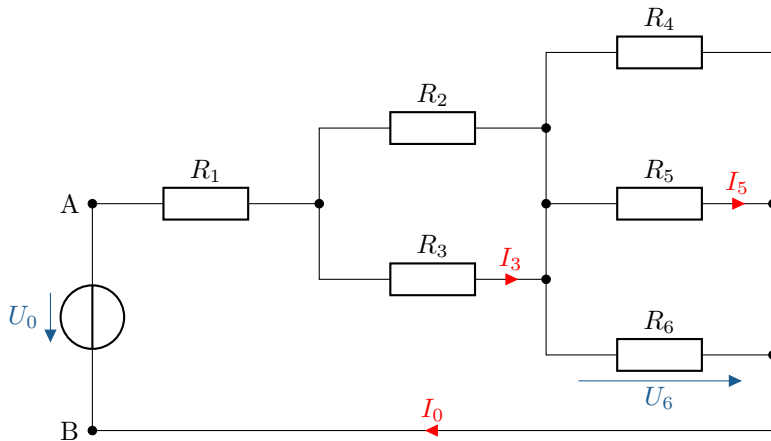
b)



A.2 Kirchhoff's laws 2

Calculate the total resistance R_{tot} between points A and B, the currents I_0 , I_3 , I_5 and the voltage U_6 for the circuit. The following values are given:

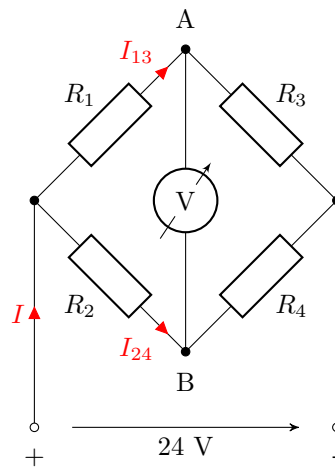
$$R_1 = 5 \text{ } \Omega; R_2 = 10 \text{ } \Omega; R_3 = 15 \text{ } \Omega; R_4 = 20 \text{ } \Omega; R_5 = 25 \text{ } \Omega; R_6 = 40 \text{ } \Omega; U_0 = 80 \text{ V}.$$



A.3 Kirchhoff's laws 3

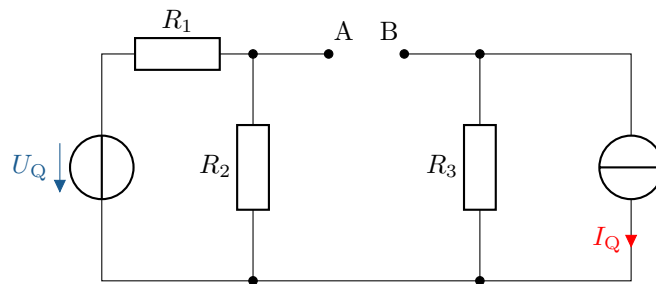
The following values are given for the bridge circuit: $R_1 = 4 \text{ k}\Omega$, $R_2 = 20 \text{ k}\Omega$, $R_3 = 6 \text{ k}\Omega$, $R_4 = 20 \text{ k}\Omega$. The internal resistance of the voltmeter can be assumed to be infinitely large. Calculate the following quantities:

- I_{13} and I_{24}
- U_1 , U_2 , U_3 and U_4
- The voltage between points A and B.
- When would the bridge be balanced, i.e. when is the voltage between points A and B zero? Express the balancing condition in as general a form as possible as a function of resistors R_1 to R_4 .



A.4 Kirchhoff's laws 4

Given is the circuit shown, consisting of the three resistors R_1 , R_2 and R_3 and the two sources U_Q and I_Q .



- Determine the internal resistance of the circuit with respect to terminals A and B.
- Calculate the open-circuit voltage U_{ABL} between terminals A and B.
- Specify the equivalent voltage source for the circuit shown in the figure above and label the quantities.

- d) What power loss P_V is converted in the circuit shown in the figure above?
- e) Terminals A and B are short-circuited. What current I_K flows through this connection between terminals A and B?
- f) What voltage U_{AB} is established when a resistor R is connected to terminals A and B? ($R_1 = R_2 = R_3 = R$)

B Solutions to exercise tasks

B.1 Kirchhoff's laws

Hier entsteht eine Musterlösung...

B.2 Kirchhoff's laws 2

Hier entsteht eine Musterlösung...

B.3 Kirchhoff's laws 3

Hier entsteht eine Musterlösung...

B.4 Kirchhoff's laws 4

Hier entsteht eine Musterlösung...

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